

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Equinix SV15, CA -- station KSJC, America/Los_Angeles
Committed pair OSM 37877544 -> 271893323
Lumen Sunnyvale 1 / Csquare SF03
Facade gap 100.5 m between the two halls
Weather record 43,747 real hourly records from KSJC
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.2925 C at 300 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-10-22 (chatter)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 1 mode change. That is 4 more than the reactive incumbent, at 0 unsafe hours against its 2. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 7 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 3 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	17.341	18.0	17.220	0.883
01	mechanical	yes	switch budget	17.402	18.0	17.220	0.858
02	mechanical	yes	switch budget	17.890	18.0	18.330	0.692
03	mechanical	yes	switch budget	17.679	18.0	17.780	0.670
04	mechanical	no	dry-bulb	18.175	18.0	18.330	0.826
05	mechanical	no	dry-bulb	18.137	18.0	17.220	0.726
06	mechanical	yes	-	16.880	18.0	14.624	0.692
07	mechanical	yes	-	17.919	18.0	14.576	1.012
08	mechanical	no	dry-bulb	18.479	18.0	14.576	1.231
09	mechanical	no	dry-bulb	18.474	18.0	14.576	1.777
10	mechanical	no	dry-bulb	19.589	18.0	15.696	1.873
11	mechanical	no	dry-bulb	20.247	18.0	17.459	1.587

12	mechanical	no	dry-bulb	20.286	18.0	18.058	1.608
13	mechanical	no	dry-bulb	20.205	18.0	18.340	1.488
14	mechanical	no	dry-bulb	20.558	18.0	19.002	1.434
15	mechanical	no	dry-bulb	19.397	18.0	17.883	1.297
16	mechanical	no	dry-bulb	18.874	18.0	18.433	1.054
17	FREE	yes	-	17.919	18.0	16.712	1.232
18	FREE	yes	-	16.311	18.0	16.111	1.143
19	FREE	yes	-	16.081	18.0	15.013	1.404
20	FREE	yes	-	15.144	18.0	14.552	1.467
21	FREE	yes	-	15.196	18.0	14.453	1.463
22	FREE	yes	-	14.660	18.0	14.441	1.075
23	FREE	yes	-	14.376	18.0	14.468	0.803

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.341 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.402 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.679 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.175 C against a 18.0 C limit, so it fails by 0.175 C. It would take a limit 0.175 C higher, or a bound 0.175 C tighter, to change this hour.

05:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.137 C against a 18.0 C limit, so it fails by 0.137 C. It would take a limit 0.137 C higher, or a bound 0.137 C tighter, to change this hour.

06:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.880 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the

switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.919 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.479 C against a 18.0 C limit, so it fails by 0.479 C. It would take a limit 0.479 C higher, or a bound 0.479 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.474 C against a 18.0 C limit, so it fails by 0.474 C. It would take a limit 0.474 C higher, or a bound 0.474 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.589 C against a 18.0 C limit, so it fails by 1.589 C. It would take a limit 1.589 C higher, or a bound 1.589 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.247 C against a 18.0 C limit, so it fails by 2.247 C. It would take a limit 2.247 C higher, or a bound 2.247 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.286 C against a 18.0 C limit, so it fails by 2.286 C. It would take a limit 2.286 C higher, or a bound 2.286 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.205 C against a 18.0 C limit, so it fails by 2.205 C. It would take a limit 2.205 C higher, or a bound 2.205 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.558 C against a 18.0 C limit, so it fails by 2.558 C. It would take a limit 2.558 C higher, or a bound 2.558 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.397 C against a 18.0 C limit, so it fails by 1.397 C. It would take a limit 1.397 C higher, or a bound 1.397 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.874 C against a 18.0 C limit, so it fails by 0.874 C. It would take a limit 0.874 C higher, or a bound 0.874 C tighter, to change this hour.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.919 C, which is 0.081 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.232 C, and the margin is measured, not chosen: 0.955 C of group-conditional forecast error for this hour of day, plus 0.2768 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.311 C, which is 1.689 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.143 C, and

the margin is measured, not chosen: 0.943 C of group-conditional forecast error for this hour of day, plus 0.2009 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.081 C, which is 1.919 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.404 C, and the margin is measured, not chosen: 1.163 C of group-conditional forecast error for this hour of day, plus 0.2410 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.144 C, which is 2.856 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.467 C, and the margin is measured, not chosen: 1.323 C of group-conditional forecast error for this hour of day, plus 0.1440 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.196 C, which is 2.804 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.463 C, and the margin is measured, not chosen: 1.262 C of group-conditional forecast error for this hour of day, plus 0.2009 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.660 C, which is 3.340 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.075 C, and the margin is measured, not chosen: 0.860 C of group-conditional forecast error for this hour of day, plus 0.2150 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.376 C, which is 3.624 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.803 C, and the margin is measured, not chosen: 0.608 C of group-conditional forecast error for this hour of day, plus 0.1948 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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